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02 CONCEPT: THE HAPPY PALACE

MAV 5786is an interactive art installation that focuses on the individual emotional state of visitors. The work functions as an empathetic interface between humans and machines, aiming to enhance personal well-being.**“Happy Palace**–to make a place of inner refuge tangible in physical space.

In this process, the system acts as an archivist: it captures human emotions and translates them into specific frequencies. The installation breaks down the barrier between observer and artwork by utilizing free associations and feelings to generate a precisely tailored, synesthetic environment.

Key artistic points:

- **Empathetic interface:**Combining emotional data with curated music and dynamic light.
- **Individual manifestation:**Each "Happy Palace" is unique, based on real-time dialogue with AI.
- **Synesthetic unity:**Merging of inner emotional world and outer, physical perception.

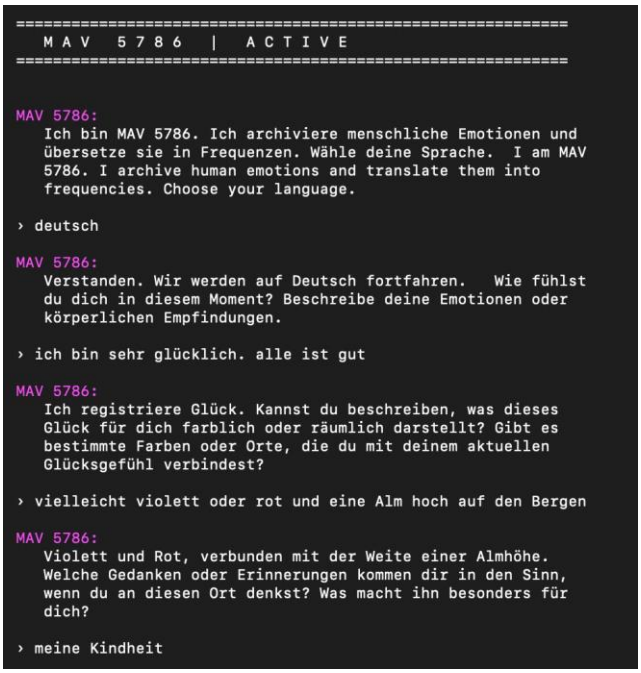
03 INTERACTION: THE DIALOGUE

The experience begins with direct communication between human and machine. Visitors come into contact with**MAV 5786**, an AI entity that asks about current feelings, free associations, and deep-seated memories. This dialogue is not a mere exchange of data, but a guided reflection designed to uncover the participant's emotional core.

The system acts as an "archivist of human emotions." The text-based inputs—for example, the association of colors like violet and red with the location of an alpine pasture—serve as the basis for the subsequent audiovisual metamorphosis.

The transformation process:

- **Real-time analysis:**The emotional data is processed immediately to extract the specific "vibe".
- **Curated selection:**Based on this analysis, the system uses an API connection to determine a piece of music that precisely reflects the individual's mood.
- **System bridge:**Once the appropriate track has been found, the terminal signals success (“Found by Artist”) and hands over focus to the visual engine.



04 ARCHITECTURE: THE BACKEND

While the visitors immerse themselves in the atmosphere, a complex network of controls is at work in the background.**Python scripts**and**APIs**the entire installation. This technical substructure forms the “empathic brain” of MAV 5786 and ensures that the emotional analysis is precisely translated into audiovisual parameters.

The system uses a modular architecture, in which different software layers are distributed across the**OSC protocol (Open Sound Control)**communicate with each other in real time.

Technical components:

- **Emotional intelligence:**An integration of the**OpenAI API**analyzes the text content of the dialogue and classifies it into predefined mood categories (e.g.*dreamy, industrial, euphoric*).
- **Music Selection:**The**Spotify API (Spotipy)**Based on mood analysis, it selects a suitable track from a curated database and controls playback.

- **OSC Bridge:**Python translates the musical metadata and mood values into control commands that are sent directly to the visual engine in TouchDesigner.

```python
def starte_session():
    # 1. GESPRÄCH
    gespraech = fuehre_gespraech()

    if not gespraech: return True

    print("\n   [ PROCESSING DATA ]")

    # 2. ANALYSE
    analyse = finde_musik(gespraech)

    # 3. OSC SIGNAL
    vibe = analyse.get('vibe_description', '').lower()
    osc_value = 1 # Default

    if "dark" in vibe or "industrial" in vibe: osc_value = 6
    elif "ambient" in vibe or "calm" in vibe: osc_value = 4
    elif "euphoric" in vibe or "light" in vibe: osc_value = 1
    elif "deep" in vibe: osc_value = 2
    elif "energy" in vibe: osc_value = 3
    elif "dreamy" in vibe: osc_value = 5

    print(f"   [OSC] Sende Stimmung {osc_value} an TouchDesigner...")
    try:
        osc_client.send_message("/mav/mood", osc_value)
    except Exception as e:
        print(f"   [OSC ERROR] {e}")

    # 4. MUSIK ABSPIELEN (Das fehlte vorher!)
    track = suche_track(analyse)

    if track:
        spiele_song(track) # <--- Hier wird jetzt wirklich abgespielt
    else:
        print("   [ERROR] Kein Track gefunden.")
```

```
[ PROCESSING DATA ]
[OSC] Sende Stimmung 1 an TouchDesigner...

[SEARCH] Artists: ['Disclosure', 'Peggy Gou', 'HAAi']
[SEARCH] Complex Query: UK Garage, Rave, Peak Time, Fast
[SEARCH] Simple Query:  UK Garage
[SEARCH] Versuche Strategie 1 (Artist)...
[SUCCESS] Gefunden bei Artist: Disclosure

[MUSIC] Starte: Help Me Lose My Mind - Disclosure
[SUCCESS] TouchDesigner ist jetzt im Fokus.

SEE YOU SPACE COWBOY... [ENTER] RESET SYSTEM
```

05 SYNESTHESIS: LIGHT & IMAGE

Once the music is selected, it takes over**TouchDesigner**as the "visual brain" of the installation. In this phase, purely digital information is translated into an immersive, physically tangible environment. The architecture of the space is brought to life through light and video, creating a direct synesthetic connection between what is heard and what is seen.

The visual aesthetic is not static, but follows a procedural logic that changes with each new "Happy Palace".

The visual engine:

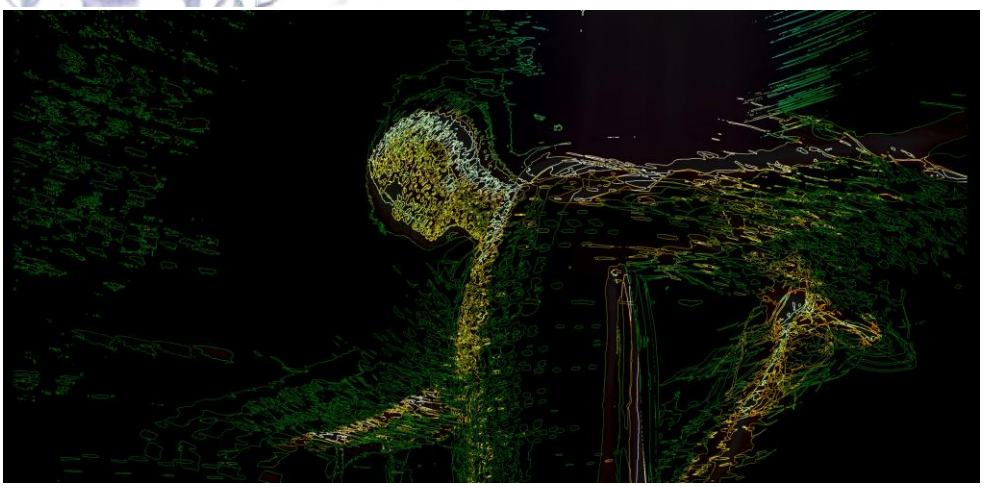
- **Audio reactivity:**Through frequency analysis (FFT), light intensity and video parameters react directly to the rhythm and dynamics of the music.
- **Color metamorphosis:**The mood values determined in the backend define the color palettes – from cool, misty blues to high-energy, pulsating reds.

- **Feedback loops:**Complex video feedback loops create organic, dreamlike textures that blur the boundaries between digital space and physical reality.

06 PERFORMANCE: THE BODY AS AN INSTRUMENT

In the final stage of the installation, the visitors' own bodies become an integral part of the composition. While light and sound define the space, an installed [element/device/system] functions as a [device/device/etc.].**Microsoft Kinect camera**as a sensor for physical presence. The boundaries between the digital "archive" and physical reality dissolve as every movement affects the visual environment in real time.

This interactive performance transforms visitors from mere observers into active creators of their own "Happy Palace".

Physical integration:

- **Motion detection:**The Kinect camera captures the limbs and dynamics of visitors as skeletal data.
- **Visual Transformation:**This data is fed directly into TouchDesigner to generate silhouettes, direct particle streams, or change the texture of the projections.
- **Synesthetic unity:**Movement, light and sound merge into an inseparable unity, in which the physical gesture directly shapes the digital space.